PULLBACK OF A LINE BUNDLE ALONG MULTIPLICATION BY N ON AN ABELIAN VARIETY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-apply-cube (canonical tag 0BFE). cited at proof lines 1957-1962 (base) and 1969-1976 (induction step). The cube identity pulled back along two diagonals.

Source reference. Original lemma 0BFE: sources/stacks-chow/tags.txt, source lines 14382–14382. Cited at proof lines 1957-1962 and 1969-1976 (pulled back twice); load-bearing.

Lemma 0.2. *Let k be a field. Let A be an abelian variety over k. Let $\mathcal{L}$ be an invertible $\mathcal{O}_A$-module. Then*

$$[n]^*\mathcal{L} \cong \mathcal{L}^{\otimes n(n+1)/2} \otimes ([-1]^*\mathcal{L})^{\otimes n(n-1)/2}$$

where $[n] : A \to A$ sends x to $x + x + \ldots + x$ with n summands and where $[-1] : A \to A$ is the inverse of A.

Proof. Consider the morphism $A \to A \times_k A \times_k A$, $x \mapsto (x, x, -x)$ where $-x = [-1](x)$. Pulling back the relation of Lemma 0BFE we obtain

$$\mathcal{L} \otimes \mathcal{L} \otimes \mathcal{L} \otimes [-1]^*\mathcal{L} \cong [2]^*\mathcal{L}$$

which proves the result for $n = 2$. By induction assume the result holds for $1, 2, \ldots, n$. Then consider the morphism $A \to A \times_k A \times_k A$, $x \mapsto (x, x, [n-1]x)$. Pulling back the relation of Lemma 0BFE we obtain

$$[n+1]^*\mathcal{L} \otimes \mathcal{L} \otimes \mathcal{L} \otimes [n-1]^*\mathcal{L} \cong [2]^*\mathcal{L} \otimes [n]^*\mathcal{L} \otimes [n]^*\mathcal{L}$$

and the result follows by elementary arithmetic. □

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